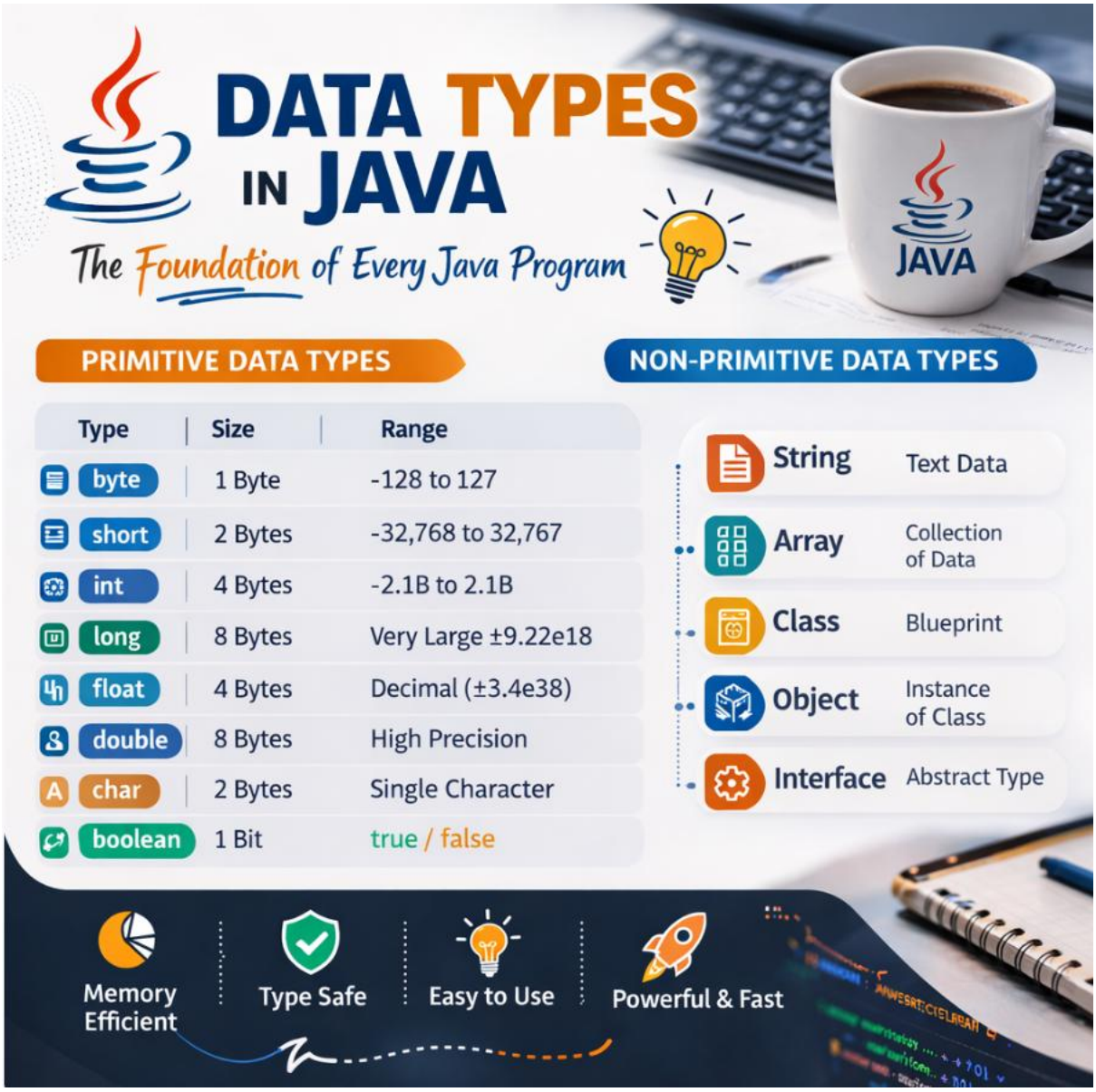
DATA TYPES IN JAVA



📊 Data Types in Java :

In Java, data types specify the type of values a variable can hold. They define the size, range and nature of data stored in memory. Java has two main categories of data types :

1.Primitive data types : byte, short, int, long, float, double, char, boolean.

2.Non-Primitive data types: String, Arrays, Classes, Interfaces, Objects.

```
Java Data Types
│
├── Primitive Data Types
│    ├ byte
│    ├ short
│    ├ int
│    ├ long
│    ├ float
│    ├ double
│    ├ char
│    └ boolean
│
└── Non-Primitive Data Types
     ├ String
     ├ Array
     ├ Class
     ├ Interface
     └ Object
```

🔹 1️⃣ Primitive Data Types :

Primitive data types are the basic built-in data types in Java.

They store actual values directly in memory and are very fast and memory efficient.

Java has 8 primitive data types.

✓ stored directly in memory
✓ fixed memory size
✓ very fast
✓ not objects

Java has 8 primitive data types

byte

short

int

long

float

double

char

boolean

1.byte

Meaning

byte is used to store very small integer numbers.

Memory Size

1 byte = 8 bits

Range

-128 to 127

Example

byte age = 25;

Memory Explanation

A byte uses 8 bits

00000000

Maximum positive value:

01111111 = 127

Minimum value:

10000000 = -128

Use Case

Used in

- file handling
- network streams
- large arrays to save memory

2.short

Meaning

short stores small integer numbers but larger than byte.

Memory Size

2 bytes = 16 bits

Range

-32,768 to 32,767

Example

short temperature = 200;

Use Case

Used when numbers are not too large but bigger than byte.

Example:

- small counters
- sensor values

3.int

Meaning

int is the most commonly used integer type in Java.

Memory Size

4 bytes = 32 bits

Range

-2,147,483,648 to 2,147,483,647

Example

int salary = 50000;

Why int is used most

Because it balances:

- memory usage
- performance
- range

Use Cases

- loops
- counters
- calculations
- indexes

Example:

```
for(int i = 0; i < 10; i++)
{
   System.out.println(i);
}
```

4.long

Meaning

long is used for very large integer numbers.

Memory Size

8 bytes = 64 bits

Example

long population = 8000000000L;

Note:

L is added to tell Java that the number is long.

Use Cases

- world population
- timestamps
- bank transactions
- large IDs

5.float

Meaning

float stores decimal numbers (single precision).

Memory Size

4 bytes

Example

float price = 19.99f;

Note:

f is required because decimal numbers are double by default in Java.

Example Values

3.14
12.5
0.75

Use Cases

- graphics
- measurements
- scientific calculations

6. double

Meaning

double stores decimal numbers with higher precision than float.

Memory Size

8 bytes

Example

double pi = 3.14159265359;

Why double is preferred

Because it provides more accuracy.

Use Cases

- financial calculations
- scientific data
- precise mathematics

7.char

Meaning

char stores a single character.

Memory Size

2 bytes

Java uses Unicode for characters.

Example

char grade = 'A';

Important:

Character must be inside single quotes

'A'
'B'
'9'
'@'

Unicode Example

'A' = 65
'B' = 66

Use Cases

- text processing
- storing letters
- symbol representation

8. Boolean

Meaning

boolean stores true or false values.

Example

```
boolean isJavaFun = true;
```

Possible values

true
false

Use Cases

Used in decision making

Example:

```
boolean isLoggedIn = true;

if(isLoggedIn)
{
   System.out.println("Welcome User");
}
```

- 2. Non-Primitive Data Types

Non-primitive types are also called Reference Data Types.

Instead of storing the value directly, they store the memory address (reference).

Examples:

String
Array
Class
Interface
Object

1. String

Used to store text data.

Example:

String name = "Omkar";

Strings are actually objects in Java.

2.Array

An array stores multiple values of the same type.

Example:

```
int numbers[] = {10,20,30,40};
```

Memory view:

```
numbers
 |
 |---- 10
 |---- 20
 |---- 30
 |---- 40
```

3. Class

A class is a blueprint to create objects.

Example:

```
class Student
{
   int age;
}
```

4.Object

Object is an instance of a class.

Example:

```
Student s1 = new Student();
```

Here

Student → class
s1 → object

Exploring Java Data Types Through a Simple Demo Application : -

```
public class DataTypesDemo {
   public static void main(String[] args) {

      // -------- Primitive Data Types --------
      byte b = 100;           // 1 byte
      short s = 30000;        // 2 bytes
      int i = 100000;         // 4 bytes
      long l = 10000000000L;    // 8 bytes

      float f = 3.14f;        // 4 bytes
      double d = 3.14159265359;  // 8 bytes

      char c = 'A';           // 2 bytes (Unicode character)
      boolean flag = true;      // 1 bit

      // -------- Non-Primitive Data Types --------
      String str = "Hello, Java"; // String (class in Java)
      int[] arr = {1, 2, 3, 4, 5}; // Array
      Integer wrapperInt = Integer.valueOf(50); // Wrapper class example
      StringBuilder sb = new StringBuilder("Java"); // Class object

      // -------- Output --------
      System.out.println("byte: " + b);
      System.out.println("short: " + s);
      System.out.println("int: " + i);
      System.out.println("long: " + l);
```

```
        System.out.println("float: " + f);
        System.out.println("double: " + d);
        System.out.println("char: " + c);
        System.out.println("boolean: " + flag);

        System.out.println("String: " + str);
        System.out.print("Array: ");
        for (int num : arr) {
            System.out.print(num + " ");
        }
        System.out.println();

        System.out.println("Wrapper Integer: " + wrapperInt);
        System.out.println("StringBuilder: " + sb);
    }
}
```

Difference between primitive and non -primitive Data types ?

Feature	Primitive Data Types	Non-Primitive Data Types
Definition	Basic built-in data types provided by Java	Complex data types created using classes
Data Stored	Stores actual value directly	Stores reference (memory address) of the object
Memory Size	Fixed memory size	Memory size can vary
Performance	Faster because value is stored directly	Slightly slower due to reference handling
Number of Types	Only 8 types	Unlimited types
Created By	Predefined by Java language	Created by programmers or Java libraries
Methods	Do not have methods	Can have methods and properties
Null Value	Cannot store null	Can store null
Example Types	byte, short, int, long, float, double, char, boolean	String, Array, Class, Object, Interface
Storage Location	Usually stored in stack memory	Stored in heap memory